MG Week 2: Assignment 2.1: Problem Domain Black Box: Home Weather Station

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1 Raspberry Pi Weather Station

Your system of interest is a home weather station. Use the following site as a reference for your weather station. https://projects.raspberrypi.org/en/projects/build-your-own-weather-station/1. You are free to enhance and modify your weather station but it must satisfy the core purpose described by the raspberrypi site. e.g. you can add an additional sensor, swap one sensor type for another or use a different communication technology.

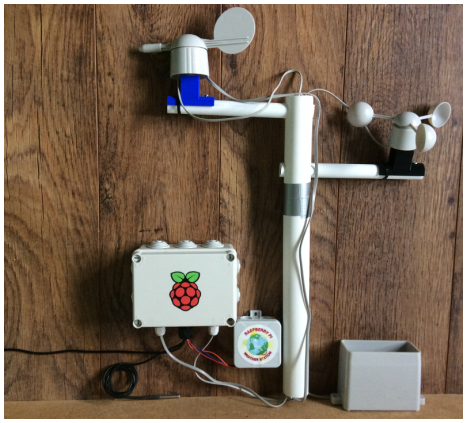

Figure 1: Raspberry Pi Weather Station

In addition to the hardware and embedded software listed above your system must also include weather monitoring software to run a device of your choice. e.g. smart phone, laptop, etc.

For measures of effectiveness include communication range (wifi) and weather reading archiving.

2 Problem Domain Black Box

Black-Box perspective

Problem domain analysis and definition from the black-box perspective is the first step towards specifying what the SoI shall do to fulfill the stakeholders' expectations. The goal of this phase is to understand how the SoI interacts with its environment to solve the given challenges. As the SoI is considered a black box, the analysis is not concerned with its internal structure and behavior. The analysis is performed to define the main inputs and outputs, black-box functions, and quantifiable characteristics of the SoI, while operating in a variety of system contexts [PHD21].

			Pillar				
			Requirements	Structure	Behavior	Parameters	Safety & Reliability
Domain	Problem	Black Box	Stakeholder Needs	System Context	Use Cases	Measures of Effectiveness (MoEs)	Conceptual and Functional Failure Mode & Effects Analysis (FMEA)
		White Box		Conceptual Subsystems	Functional Analysis	MoEs for Subsystems	Conceptual Subsystems FMEA
	Solution		System Requirements	System Structure	System Behavior	System Parameters	System Safety & Reliability (S&R)
			Subsystem Requirements	Subsystem Structure	Subsystem Behavior	Subsystem Parameters	Subsystem S&R
			Component Requirements	Component Structure	Component Behavior	Component Parameters	Component S&R
	Implementa-tion		Implementation Requirements				

Figure 2: MagicGrid® Black Box

Use the MagicGrid BOK as a reference.

Tasks:

1. MBSE Planning
2. Stakeholder Needs
   (a) Capture information directly in the model (MG BOK Sub-Step 3)
   (b) Create 5 to 10 requirements for your Weather Station
3. System Context
4. Use Cases
   (a) Add item flows as object flows to scenarios (MG BOK Sub-Step 7)

5. Measures of Effectiveness
6. Review

Outputs:

1. Project in TWC
2. Table of stakeholder needs
3. System context diagram in BDD, IBD
4. Use cases identified in UC
5. Scenarios defined in ACTs
6. MoEs defined in BDD
7. Review captured in model

References

[PHD21] Aurelijus Morkevičius PH.D. *MagicGrid® BOOK OF KNOWLEDGE*. A Practical Guide to Systems Modeling using MagicGrid from Dassault Systèmes. VITAE LITERA, 2021. ISBN: 978-609-454-554-2.